

Grid Search: the four display statistics

pyHYSCORE / `grid_search.py`

The grid search scans two spin-Hamiltonian parameters and, at every grid point, simulates each active dataset and compares it with experiment. The **Display** pulldown chooses which of four statistics is mapped. This note defines each one, explains what it does and does not tell you, and gives references for the last three.

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1 Notation

Indices and per-dataset quantities

Symbol	Meaning
$\theta = (\theta_1, \theta_2)$	the pair of scanned spin-Hamiltonian parameters; one grid point
k	index over active (shown) datasets
D	the data: all active datasets taken together
e_k	experimental spectrum of dataset k , restricted to its region of interest, as a vector
$s_k(\theta)$	simulation for dataset k at θ , interpolated onto the same points, as a vector
N_k	number of spectrum points in dataset k 's region of interest
$f_{\min}, f_{\max}$	frequency limits of that region, the same on both axes
$c_k(\theta)$	best-fit non-negative amplitude applied to s_k , Eq. (1)
$\text{SSE}_k(\theta)$	residual sum of squares for dataset k , Eq. (2)
null_k	null-model sum of squares $\ e_k\ ^2$: the residual left by no simulation
r_k	uncentred correlation between e_k and s_k , Eq. (7)
σ_k	unknown noise level of dataset k ; a nuisance parameter, marginalized away
N_k^{eff}	effective (independent) point count, N_k/F
a	an arbitrary rescaling factor applied to a dataset, used only in invariance arguments

Aggregate and displayed quantities

Symbol	Meaning
N	pooled point count $\sum_k N_k$
$\text{RMSD}(\theta)$	display 1: pooled root-mean-square residual, Eq. (3)
$\text{RMSD}_{\text{norm}}(\theta)$	display 2: null-normalized score in $(0, 1]$, Eq. (5)
$p(\theta D)$	posterior probability density over the scanned parameters, Eq. (10)
$\Phi(\theta)$	negative log posterior, the quantity actually minimized, Eq. (11)
i, j	indices of a cell on the interpolated (fine) display grid
a, b	dummy cell indices inside sums over the grid
p_{ij}	display 4: posterior normalized over the grid, $\sum_{ij} p_{ij} = 1$, Eq. (13)
L_{ij}	display 3: credible level of a cell, Eq. (14)
$p_i^{(1)}, p_j^{(2)}$	marginal distributions along each scanned axis
F	oversampling: spectrum points per independent measurement, $F = 4^z$
z	zero-fill setting from the main window
$\text{SSE}_{\min}, \text{RMSD}_{\min}$	smallest value of that quantity over the scanned grid
$F_{p, N-p}(\text{conf})$	quantile of the F distribution with p and $N - p$ degrees of freedom
$\chi_p^2(\text{conf})$	quantile of the chi-square distribution with p degrees of freedom
conf	the confidence or credible fraction, here 0.68 or 0.95

One collision to be aware of. p carries two standard meanings in this document, kept because both are conventional in their own literature: as a function of arguments, $p(\cdot)$ is always a probability density; as a bare symbol in Section 6 it is the number of jointly scanned parameters, which here is always 2. The subscripted p_{ij} , $p_i^{(1)}$, $p_j^{(2)}$ are probabilities.

Operators and abbreviations

Symbol	Meaning
$\langle x, y \rangle$	Euclidean inner product $\sum_n x_n y_n$ over the region of interest
$\ x\ $	Euclidean norm, $\ x\ ^2 = \langle x, x \rangle$
$\lceil x \rceil$	ceiling: the smallest integer not less than x
$\max(x, 0)$	clamp to zero: x if positive, otherwise 0
RMSD	root-mean-square deviation
SSE	sum of squared residuals
roi	region of interest: the $[f_{\min}, f_{\max}]^2$ window a fit is evaluated over
HPD	highest posterior density; a credible region containing the most probable values
FFT	fast Fourier transform
HYSCORE	hyperfine sublevel correlation spectroscopy

2 What one grid point produces

Write $\theta = (\theta_1, \theta_2)$ for the pair of scanned parameters. For each active dataset k let e_k be the experimental spectrum restricted to that dataset's region of interest $[f_{\min}, f_{\max}]^2$, containing N_k points, and let $s_k(\theta)$ be the simulation interpolated onto the same grid. Both are treated as vectors in $\mathbb{R}^{N_k}$.

A simulated HYSCORE spectrum carries no meaningful absolute intensity, so each dataset is given its own amplitude, fitted in closed form by least squares and clamped to be non-negative:

$$c_k(\theta) = \max\left(\frac{\langle e_k, s_k(\theta) \rangle}{\|s_k(\theta)\|^2}, 0\right). \quad (1)$$

The clamp matters later: a negative amplitude would mean an inverted simulation fits best, which is unphysical here.

The residual sum of squares for that dataset is then

$$\text{SSE}_k(\theta) = \|e_k - c_k(\theta) s_k(\theta)\|^2 = \sum_{\text{roi}} (e_k - c_k s_k)^2. \quad (2)$$

Everything below is a different way of combining the set $\{\text{SSE}_k, N_k\}$ into one number per grid point.

3 Display 1 — RMSD

$$\text{RMSD}(\theta) = \sqrt{\frac{\sum_k \text{SSE}_k(\theta)}{\sum_k N_k}}. \quad (3)$$

This is the pooled root-mean-square residual: all residuals from all datasets are thrown into one pool and the root mean square is taken. It is the original fitness measure and remains the default.

What it means. A distance in experimental intensity units. Because those units are arbitrary (spectrometer gain, number of scans), the absolute value carries no information — only comparisons between grid points within one scan are meaningful.

The property to be aware of. Equation (3) weights datasets by their absolute intensity. If dataset B is $250\times$ brighter than dataset A , its residuals are $250\times$ larger and its contribution to

the sum is about $62500\times$ larger: B effectively decides the fit alone. This is a modelling choice, not a defect, but it is rarely the intended one when several measurements of comparable quality are combined.

4 Display 2 — Normalized RMSD

Define each dataset's *null model*: the residual left by no simulation at all,

$$\text{null}_k = \|e_k\|^2 = \sum_{\text{roi}} e_k^2. \quad (4)$$

The displayed statistic is the point-count-weighted mean of the per-dataset unexplained fractions:

$$\text{RMSD}_{\text{norm}}(\theta) = \sqrt{\frac{\sum_k N_k (\text{SSE}_k(\theta) / \text{null}_k)}{\sum_k N_k}}. \quad (5)$$

4.1 Why it is bounded

Setting $c_k = 0$ in (2) gives $\text{SSE}_k = \|e_k\|^2 = \text{null}_k$ exactly. Since (1) clamps $c_k \geq 0$, the fitted amplitude can never do *worse* than $c_k = 0$, so

$$0 < \frac{\text{SSE}_k}{\text{null}_k} \leq 1 \quad \implies \quad 0 < \text{RMSD}_{\text{norm}} \leq 1. \quad (6)$$

This is a hard bound, not an empirical observation. The interpretation is absolute: 0 is a perfect fit, 1 means the simulation explains nothing.

4.2 Geometric reading

When $\langle e_k, s_k \rangle > 0$, substituting the optimum (1) into (2) gives $\text{SSE}_k = \|e_k\|^2 - \langle e_k, s_k \rangle^2 / \|s_k\|^2$, hence

$$\sqrt{\frac{\text{SSE}_k}{\text{null}_k}} = \sqrt{1 - r_k^2}, \quad r_k = \frac{\langle e_k, s_k \rangle}{\|e_k\| \|s_k\|}, \quad (7)$$

where r_k is the uncentred correlation coefficient. The normalized value is therefore the sine of the angle between experiment and simulation viewed as vectors, and r_k^2 is the fraction of experimental power the simulation accounts for. A displayed value of 0.3 means $1 - 0.3^2 = 91\%$ of the power is explained.

4.3 Brightness invariance, and its cost

Rescaling a dataset, $e_k \rightarrow a e_k$, multiplies both SSE_k and null_k by a^2 , leaving the ratio unchanged. Every dataset therefore contributes on equal terms regardless of signal strength.

The cost is that (5) is a genuinely different statistic from (3): it is a weighted sum of SSE_k with weights N_k / null_k rather than uniform weights. With datasets of unequal brightness the two surfaces have different shapes, and the best-fit point can move between them. Neither is the display being wrong — they answer different questions about how much a faint dataset should count.

4.4 Caveat

The normalization is uncentred: no baseline is subtracted. Magnitude spectra have a positive noise floor, which inflates null_k and flattens the score. Two datasets with different baselines are therefore not strictly comparable on this scale.

5 Displays 3 and 4 — posterior probability

5.1 From residuals to a probability

Assume the residuals of dataset k are Gaussian with an unknown noise level σ_k , independently for each dataset. The likelihood is

$$p(D \mid \theta, \{\sigma_k\}) \propto \prod_k \sigma_k^{-N_k} \exp\left(-\frac{\text{SSE}_k(\theta)}{2\sigma_k^2}\right). \quad (8)$$

The noise levels are nuisance parameters. Assigning each the scale-invariant Jeffreys prior $p(\sigma_k) \propto 1/\sigma_k$ and integrating,

$$\int_0^\infty \sigma^{-N-1} \exp\left(-\frac{\text{SSE}}{2\sigma^2}\right) d\sigma \propto \text{SSE}^{-N/2}, \quad (9)$$

which leaves a posterior over the scanned parameters alone:

$$\boxed{p(\theta \mid D) \propto \prod_k \text{SSE}_k(\theta)^{-N_k/2}} \quad (10)$$

(times a prior on θ , taken flat over the scanned box). Equivalently, the quantity minimized is

$$\Phi(\theta) = -\log p(\theta \mid D) = \sum_k \frac{N_k}{2} \log \text{SSE}_k(\theta) + \text{const.} \quad (11)$$

5.2 Two consequences

It is the logarithm of each dataset’s residual that gets averaged, not the residual itself. That is what marginalizing an unknown per-dataset noise level produces, and it is the principled answer to the weighting question raised in Section 4.3 — the weights are inferred rather than chosen.

It is automatically invariant to brightness and to the normalization of Section 4. Rescaling $e_k \rightarrow a e_k$ sends $\log \text{SSE}_k \rightarrow \log \text{SSE}_k + 2 \log a$, a constant shift of Φ that cancels on normalization. Likewise $\log(\text{SSE}_k/\text{null}_k) = \log \text{SSE}_k - \log \text{null}_k$ differs by a constant. The probability map is therefore identical whether or not the RMSD is normalized — the question that makes displays 1 and 2 disagree simply does not arise here.

5.3 Effective number of points

N_k enters (10) as an exponent, so it controls how sharp the posterior is; the width of a credible region scales as $N^{-1/2}$. Adjacent points of a processed spectrum are not independent: apodization correlates neighbours, and zero-filling manufactures points carrying no new information at all. The **Oversampling** control divides the point count accordingly,

$$N_k^{\text{eff}} = \frac{N_k}{F}, \quad F = 4^z = (2^z)^2, \quad (12)$$

where z is the zero-fill setting: `update_FFT` pads to $2^{\lceil \log_2 N \rceil + z}$, so each axis gains 2^z points and the two-dimensional count gains that squared. Leaving $F = 1$ overstates the information content and yields credible regions too narrow by a factor $\sqrt{F}$.

5.4 On the grid

Φ is evaluated at the coarse scan points, interpolated (the negative log posterior is smooth and roughly quadratic near its minimum, whereas the probability itself spans many orders of magnitude and interpolates badly), then exponentiated and normalized over the fine grid:

$$p_{ij} = \frac{\exp(-[\Phi_{ij} - \min \Phi])}{\sum_{ab} \exp(-[\Phi_{ab} - \min \Phi])}, \quad \sum_{ij} p_{ij} = 1. \quad (13)$$

Failed grid points get $p_{ij} = 0$. Note that (13) makes the scanned box the prior: the result is only meaningful if the posterior has decayed to negligible before the edges, which the panel checks and warns about.

5.5 Display 3 — credible level

Each cell is labelled with the total probability mass of all cells at least as probable as it:

$$L_{ij} = \sum_{\{ab: p_{ab} \geq p_{ij}\}} p_{ab} \in (0, 1]. \quad (14)$$

This is the smallest highest-posterior-density region that contains the cell. It runs from near 0 at the most probable point to 1 at the least, so contouring at 0.68 and 0.95 draws exactly the 68 % and 95 % credible regions. Cells of equal density share one level, so a ring of equally probable cells is never split across a contour.

This is the recommended view: it is readable everywhere, and its contours are credible regions by construction rather than by a threshold rule.

5.6 Display 4 — posterior density

The same object before the cumulative step: p_{ij} from (13) itself, the probability mass per grid cell. It is the primary quantity, and L is a monotone function of it, so both produce identical contours. Its drawback is visual: a sharp posterior puts nearly every cell at effectively zero and only a few pixels light up. It is most useful when the posterior is broad.

5.7 Uncertainty reported in these modes

Instead of the F-test region used for displays 1 and 2, the interval comes from the marginal distributions

$$p_i^{(1)} = \sum_j p_{ij}, \quad p_j^{(2)} = \sum_i p_{ij}, \quad (15)$$

reported as the highest-density interval containing 95 % of the marginal mass. Marginalizing is the principled version of what the bounding box of a joint region approximates, and it is generally tighter.

6 For contrast: intervals in displays 1 and 2

There the noise level is not modelled, so the confidence region comes from the ratio of sums of squares. With $p = 2$ scanned parameters, the F-test region for nonlinear least squares with unknown variance is

$$\text{SSE}(\theta) \leq \text{SSE}_{\min} \left[1 + \frac{p}{N-p} F_{p, N-p}(\text{conf}) \right], \quad (16)$$

so the boundary in RMSD is $\text{RMSD}_{\min}$ times the square root of that bracket. The alternative $\Delta\chi^2$ option uses $1 + \chi_p^2(\text{conf})/N$, which under-corrects for finite N and gives narrower regions. Both assume a plain sum-of-squares ratio, which the dataset-reweighted mean (5) only approximates — a further reason to prefer the posterior modes when several datasets are combined.

7 Summary

Display	Range	Units	Weighting of datasets
RMSD	$(0, \infty)$	experimental intensity	by absolute intensity
Normalized RMSD	$(0, 1]$	dimensionless	equal (arithmetic)
Credible level	$(0, 1]$	probability mass	inferred (logarithmic)
Posterior density	$(0, 1]$	mass per cell	inferred (logarithmic)

8 References

The last three displays have close analogues in the structural-biology and Bayesian-inference literature.

Normalized RMSD

G. Cornilescu, J. L. Marquardt, M. Ottiger, A. Bax, *Validation of protein structure from anisotropic carbonyl chemical shifts in a dilute liquid crystalline phase*, J. Am. Chem. Soc. **120**, 6836–6837 (1998).

doi:10.1021/ja9812610 <https://doi.org/10.1021/ja9812610>

Introduces the quality factor $Q = \text{rms}(D_{\text{obs}} - D_{\text{calc}})/\text{rms}(D_{\text{obs}})$, algebraically the same construction as (5): residual divided by the intrinsic scale of the data. Arrived at for the same reason — the raw number is in arbitrary units and means nothing on its own.

A. T. Brünger, *Free R value: a novel statistical quantity for assessing the accuracy of crystal structures*, Nature **355**, 472–475 (1992).

doi:10.1038/355472a0 <https://doi.org/10.1038/355472a0>

The canonical treatment of normalized residual factors, and of why a fit statistic evaluated on the data used to fit it is optimistic. The cross-validation remedy transfers directly: hold out one τ value and test the parameters found from the others.

Posterior probability (credible level and density)

G. L. Bretthorst, *Bayesian Spectrum Analysis and Parameter Estimation*, Lecture Notes in Statistics **48**, Springer (1988).

doi:10.1007/978-1-4684-9399-3 <https://doi.org/10.1007/978-1-4684-9399-3>

The closest reference of all. Derives (10) for spectral data: amplitude parameters that enter linearly are marginalized analytically, an unknown noise level is integrated out under a Jeffreys prior, and what remains is a posterior over the nonlinear parameters proportional to a negative power of the residual sum of squares. Equations (1) and (10) are that scheme applied to a two-dimensional spectrum.

E. T. Jaynes (ed. G. L. Bretthorst), *Probability Theory: The Logic of Science*, Cambridge University Press (2003).

doi:10.1017/CB09780511790423 <https://doi.org/10.1017/CB09780511790423>

The foundational treatment of nuisance-parameter marginalization and of Jeffreys priors for scale parameters, which is what turns an unknown σ_k into the exponent $-N_k/2$.

W. Rieping, M. Habeck, M. Nilges, *Inferential structure determination*, Science **309**, 303–306 (2005).

doi:10.1126/science.1110428 <https://doi.org/10.1126/science.1110428>

Applies exactly this idea to structure determination: rather than choosing weights for heterogeneous data by hand, treat the error scale of each data set as an unknown and marginalize it. This is the argument for why (11) is preferable to (5) when combining datasets.

M. Habeck, M. Nilges, W. Rieping, *Replica-exchange Monte Carlo scheme for Bayesian data analysis*, Phys. Rev. Lett. **94**, 018105 (2005).

doi:10.1103/PhysRevLett.94.018105 <https://doi.org/10.1103/PhysRevLett.94.018105>

The sampling counterpart, for when a two-parameter grid is no longer enough; also discusses inferring the weight of each data source.